Master in Computer Science
Software Engineering

Cloud Computing

Report 3

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Contents

Exercise 1. VPC Creation

Question 1. Create a VPC with CIDR Block 120.0.0.0/16

Login to the AWS Management Console. Services select VPC → Create VPC
Steps:

1. Select **VPC only** option to create VPC with customized options
2. Provide a VPC name i.e., `MyVPC1`
3. Select **IPv4 CIDR manual input** (Currently we are targeting for IPv4 only)
4. Select **Default** tenancy (Shared resources)
5. Provide the CIDR block: `120.0.0.0/16`
6. Click **Create VPC**

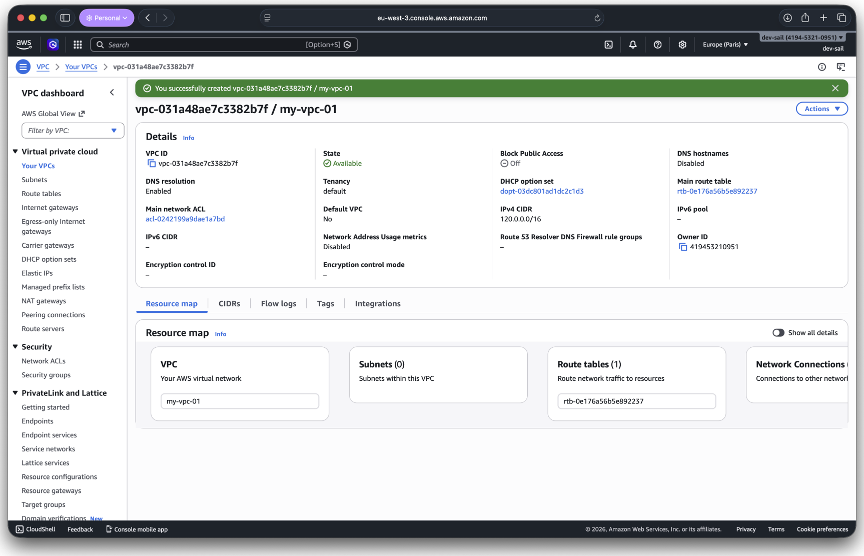

Figure 1: VPC created successfully

Exercise 2. Subnet Creation

In VPC service → Click on subnets → Create subnet

Question 1. Create Public Subnet

Steps:

1. Select the correct VPC
2. Provide a subnet Name i.e., `Public`
3. Assign the IPv4 CIDR block for this subnet `120.0.3.0/24`
4. Provide Tags for easy tracking and identification
5. Click **Create subnet**

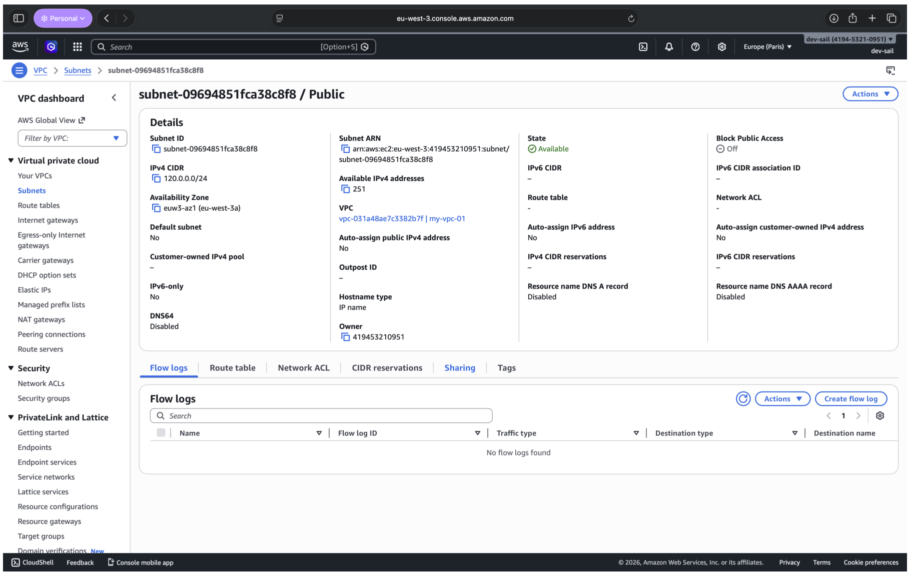

Figure 2: Public subnet created

Question 2. Create Private Subnet 1

Steps:

1. Select the correct VPC
2. Provide a subnet Name i.e., `Private1`
3. Assign the IPv4 CIDR block for this subnet `120.0.1.0/24`
4. Provide Tags for easy tracking and identification
5. Click **Create subnet**

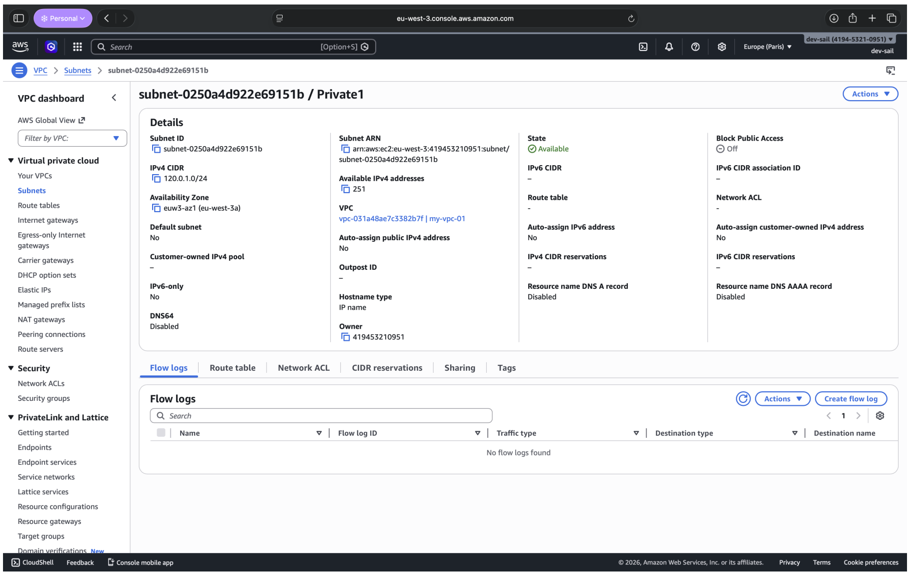

Figure 3: Private1 subnet created

Question 3. Create Private Subnet 2

Steps:

1. Select the correct VPC
2. Provide a subnet Name i.e., `Private2`
3. Assign the IPv4 CIDR block for this subnet `120.0.2.0/24`
4. Provide Tags for easy tracking and identification
5. Click **Create subnet**

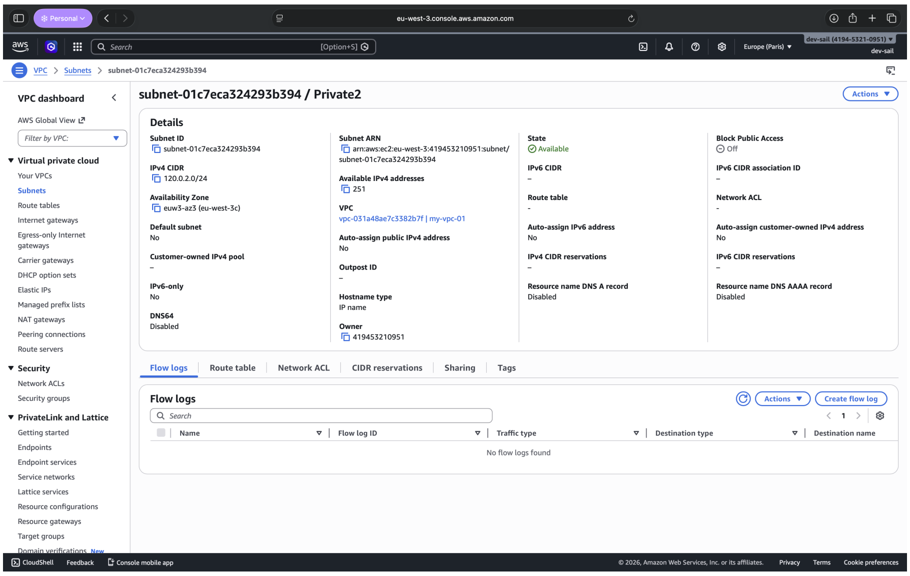

Figure 4: Private2 subnet created

Exercise 3. Internet Gateway

In VPC service → Internet gateways → Create internet gateway

Question 1. Create Internet Gateway

Steps:

1. Provide name: `igw1`
2. Click **Create internet gateway**
3. Select the created gateway and click **Actions** → **Attach to VPC**
4. Select **MyVPC1**
5. Click **Attach internet gateway**

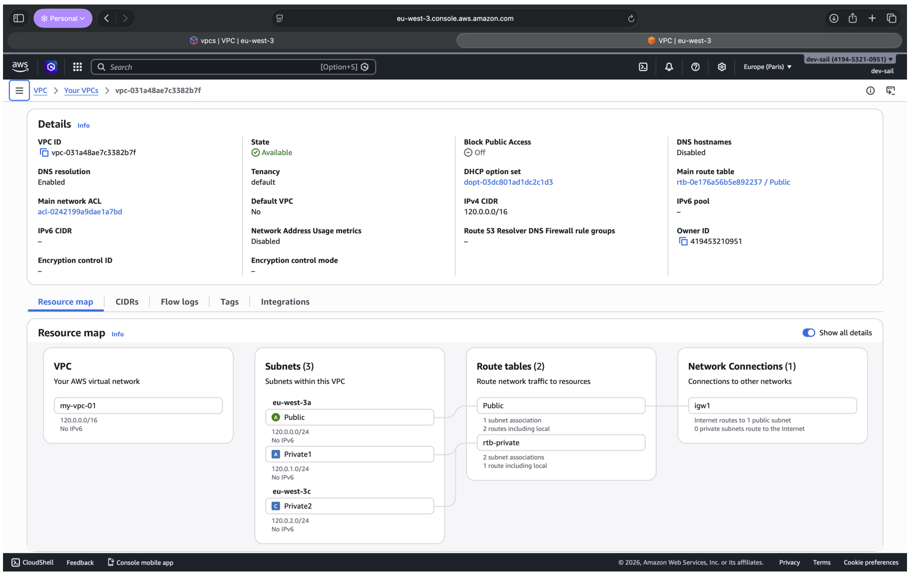

Figure 5: VPC Resource Map

Exercise 4. Route Tables

In VPC service → Route tables

Question 1. Public Route Table

Steps:

1. Navigate to Route tables
2. Edit the public route table and add the following route:
    - **Destination:** 0.0.0.0/0
    - **Target:** Internet Gateway (igw1)

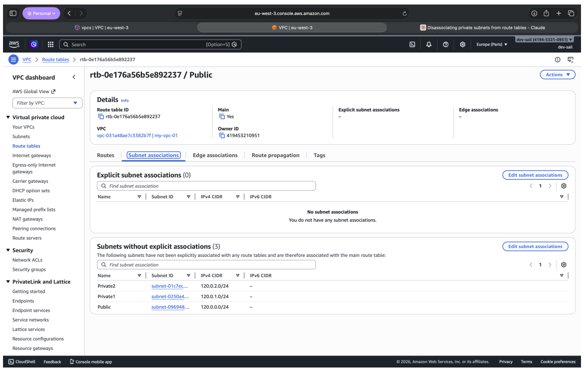

Figure 6: Public route table before

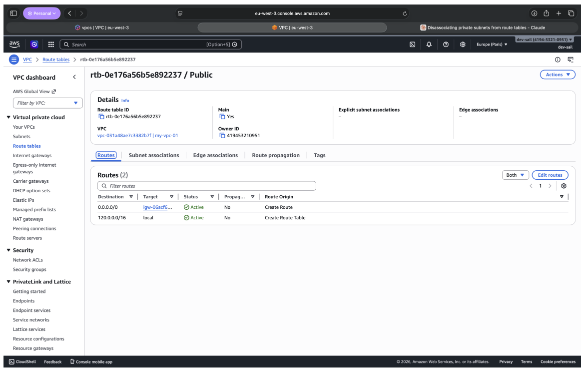

Figure 7: Adding IGW route

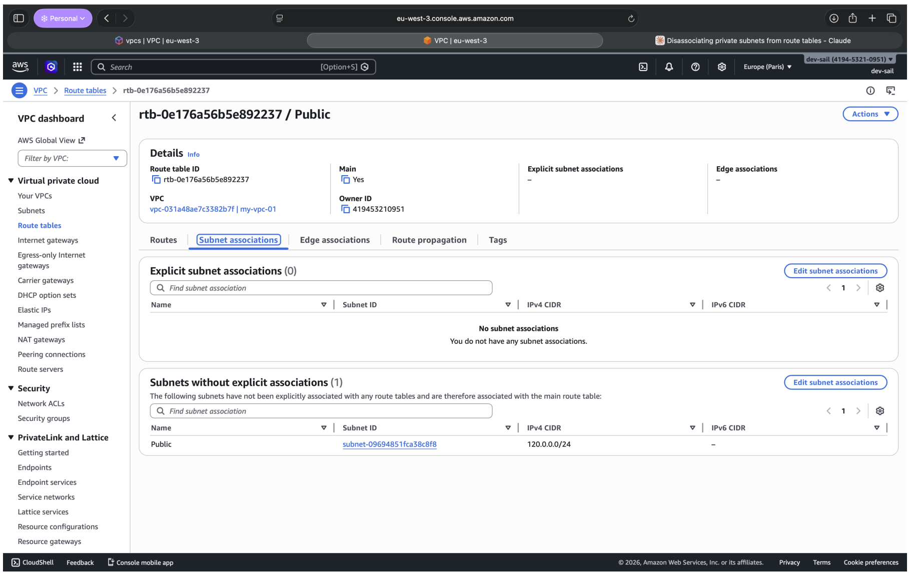

Figure 8: Public route table after adding IGW

Question 2. Private Route Table

Steps:

1. Navigate to Route tables
2. Click **Create route table**
3. Provide name: `rtb-private`

4. Select the VPC (**MyVPC1**)
5. Click **Create route table**
6. Associate the private subnets (Private1 and Private2) with this route table

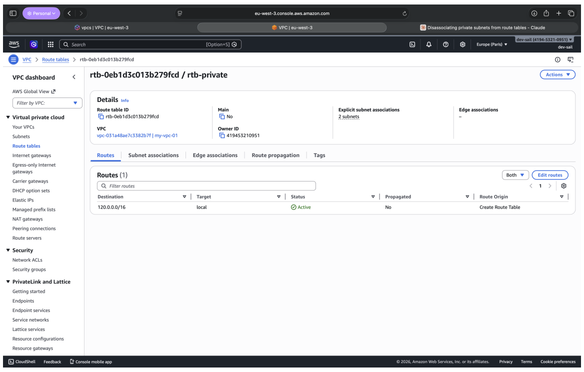

Figure 9: Private route table routes

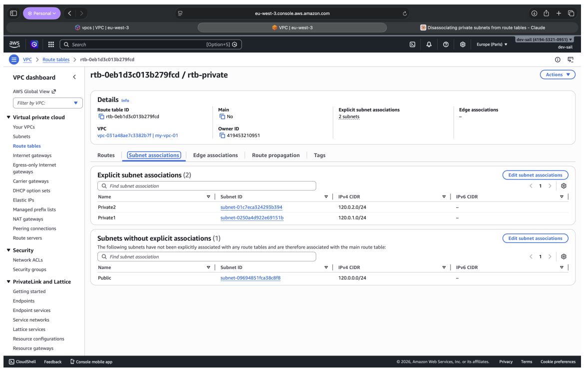

Figure 10: Private subnets association

Exercise 5. NAT Gateway

In VPC service → NAT gateways → Create NAT gateway

Question 1. Create NAT Gateway

Steps:

1. Provide name: `my-nat-gateway1`
2. Select Subnet: Public subnet
3. Select Connectivity Type: Public
4. Click **Allocate Elastic IP**
5. Click **Create NAT gateway**

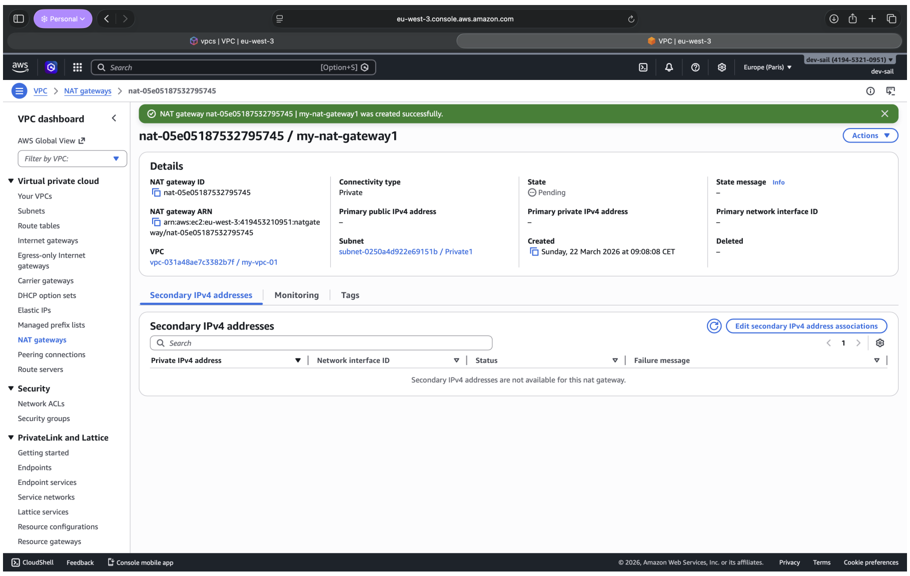

Figure 11: NAT Gateway created

Question 2. Configure Private Route Table with NAT Gateway

Steps:

1. Navigate to Route tables
2. Select the private route table (`rtb-private`)
3. Click **Edit routes**
4. Click **Add route**
5. Add the following route:
    - **Destination:** 0.0.0.0/0
    - **Target:** NAT Gateway (my-nat-gateway1)
6. Click **Save changes**

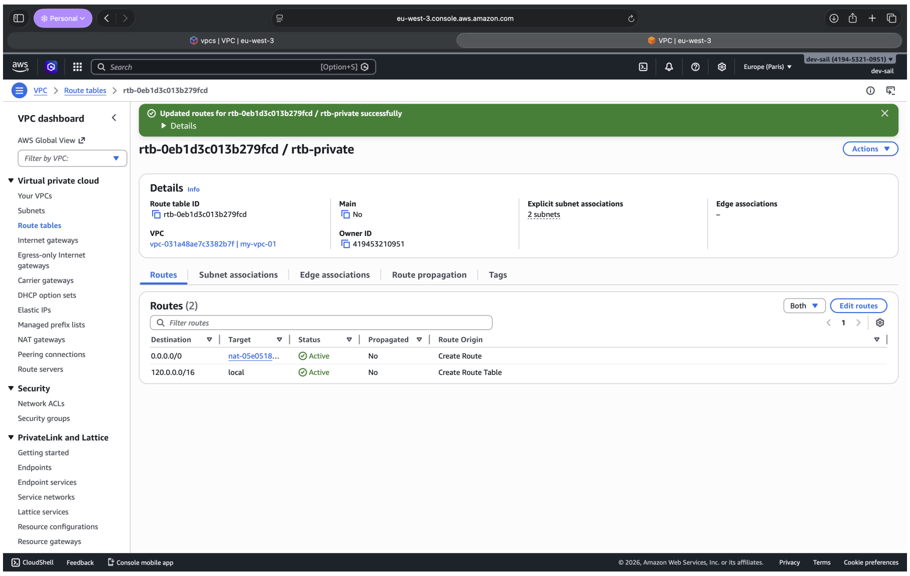

Figure 12: Private route table with NAT Gateway